

Musical Fight (trap)

Giorgio and Edoardo are talking about their favourite music genres, and the discussion is getting intense as they approach the *trap music* topic! Giorgio argues that trap lyrics are all the same, while Edoardo is a diehard fan of the genre and can't tolerate such slander. In order to prove Giorgio wrong, he decided to measure how much his favourite trap song is original.



Figure 1: Edoardo's favourite trapper, *A sphere and that's it*.

Thus, Edoardo collected the N trap greatest hits, the first of which is his favourite. For each word of his favourite song, he wants to compute in how many other songs this word has already appeared, and then come back to Giorgio with the evidence collected this way, proving *trap supremacy*!

📎 Among the attachments of this task you may find a template file `trap.*` with a sample incomplete implementation.

Input

The first line contains the two integers N and L . Further N lines follow, each describing the i -th song through an integer M_i (the number of words in the lyrics) followed by M_i space-separated words (each composed of at most L Latin lowercase letters “a” to “z” only).

Output

You need to write a single line with M_0 integers: for each word of the first song, the number of other songs in which it appears.

Constraints

- $1 \leq L \leq 10$.
- $1 \leq N \leq 1000$.
- $1 \leq M_i \leq 5000$ for each $i = 0 \dots N - 1$.
- Every word is comprised of at most L Latin lowercase letters (“a” to “z”).

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (12 points) $L = 1$.

- **Subtask 3** (14 points) $L \leq 3$.

- **Subtask 4** (27 points) $M_i \leq 100$ for each i .

- **Subtask 5** (26 points) $N \leq 100$.

- **Subtask 6** (21 points) No additional limitations.


Examples

input	output
4 9 11 oh sfera sk sk skuh marseille milano oh oui bang bang 7 sfera ebbasta oh come un bang oh 4 amo milano oh sfera 5 sfera ti bang bang ebbasta	2 3 0 0 0 0 1 2 0 2 2
5 7 14 che vi che mi avolta che qui dale merale merale dale avolta merale che 6 mi nisti nisti nisti mi mi 11 mi avolta che cele mi dale avolta cele dale cele mi 7 dale merale che che merale mi cele 8 merale merale cele avolta che dale avolta che	3 0 3 3 2 3 0 3 2 2 3 2 2 3

Explanation

In the **first sample case**, the word “sfera” appears in all other songs, while “oh” and “bang” appear in other two songs. Finally, “milano” only appears in one other song, and the other words are novel.